

Maha Sajid

Blacksburg, VA | (202) 615-0172 | mahasajid1@gmail.com | LinkedIn | Website | Google Scholar

Education

Virginia Tech -- Expected May 2028

PhD in Computer Science

Research Focus: Human-Computer Interaction, Collaborative Mixed Reality (MR), Team Cognition

Lahore University of Management Sciences (LUMS) -- Aug 2018 - May 2022

BS in Computer Science

Minor in Psychology - Graduated with Distinction

Research Experience

Graduate Research Assistant

Virginia Tech, Blacksburg, VA -- Aug 2023 - Present

Advisors: Dr. Brendan David-John and Dr. Bo Ji

Collective Situational Awareness in Asymmetric MR

- Led an end-to-end investigation of how communication modality shapes collective situational awareness in asymmetric MR, designing and implementing a networked HoloLens 2 experiment, refining the protocol through a 12-participant pilot, and conducting an IRB-approved, counterbalanced study with 48 participants across 24 dyads.
- Extended SAGAT-based collective situational awareness assessment by introducing two team-level measures--information coverage and accurate alignment--alongside established correctness and response-alignment measures, showing that aggregate scores can obscure meaningful differences in how knowledge is distributed and shared across teams.
- Built multimodal analysis pipelines integrating gaze, speech, SAGAT, workload, and performance data, deriving task-space gaze and verbal-coordination measures to characterize how teams distributed attention and coordinated across conditions.
- Found that Gaze+Voice improved task performance and shared awareness, while Gaze-only produced the strongest spatial gaze convergence, showing that performance, co-attention, and collective awareness capture distinct aspects of collaboration. Manuscript is under review at IEEE VR 2027.

Security Attacks in Remote Collaborative MR

- Designed and executed a mixed-methods security study on how users recognize and respond to attacks in remote collaborative MR, iteratively refining the task, attack conditions, and study protocol across two pilot studies with 16 participants before conducting an IRB-approved main study with 20 participants across 10 dyads.
- Engineered four real-time security attack interventions in a networked Unity-based HoloLens 2 platform, including a novel click-redirection mechanism that manipulated 3D object interactions and MR-specific occlusion attacks that exploited invisible spatial elements.
- Triangulated first-person recordings, behavioral observations, questionnaires, and semi-structured interviews to uncover gaps between users' perceived and actual security behavior; 54% reported no attack during spatial occlusion, while 67% failed to report the mitigation strategy they had actually used.
- Translated findings on attack misattribution and mitigation into design recommendations for immersive security indicators, interaction transparency, and user training; resulted in a first-author IEEE VR 2025 paper featured in Computerworld.

Bystander Privacy Indicators in Extended Reality (XR)

- Collaborated on an iterative mixed-methods study of privacy indicators for situationally impaired XR bystanders, using four focus groups with 8 participants to develop five visual and multimodal concepts followed by a 7-participant evaluation of usability and contextual usefulness.
- Analyzed qualitative and quantitative feedback across indicator designs, finding that visual-only cues were less effective under situational impairment while multimodal indicators were preferred in privacy-sensitive scenarios; the work resulted in an IEEE ISMAR Adjunct 2025 publication.

Undergraduate Research Assistant

Lahore University of Management Sciences (LUMS), Lahore, Pakistan -- Jan 2022 - Feb 2023

Advisors: Dr. Fareed Zaffar and Dr. Murtaza Taj

- Achieved 92% accuracy in identifying constructed areas by developing a satellite-imagery analysis pipeline for detecting land-use change and potentially unrecorded development, supporting prioritization of labor-intensive property-tax surveys.
- Developed a GIS-integrated decision-support platform that transformed machine-learning outputs into spatial and temporal construction comparisons, enabling users to inspect newly detected construction across locations and years.
- Integrated remote-sensing outputs into a government property-verification workflow for flagging inconsistencies between GIS and taxation records and directing field-survey resources; the work

contributed to the International Growth Centre technical report Cross Verification of Revenue Using Remote Sensing Data.

Selected Publications

- Sajid, M., Bukhari, S. I. M. S., Ji, B., and David-John, B. (2025). "Just stop doing everything for now!": Understanding Security Attacks in Remote Collaborative Mixed Reality. IEEE VR 2025. DOI: 10.1109/VR59515.2025.00085.
- Bukhari, S. I. M. S., Sajid, M., Ji, B., and David-John, B. (2025). Rethinking Privacy Indicators in Extended Reality: Multimodal Design for Situationally Impaired Bystanders. IEEE ISMAR Adjunct 2025. DOI: 10.1109/ISMAR-Adjunct68609.2025.00059.

Industry Experience

Software Engineer

Senarios, Lahore, Pakistan -- Feb 2023 - Jul 2023

- Delivered a cross-platform mobile data-sharing product from design through release, implementing real-time contact synchronization, decentralized functionality, and technical specifications for a Google Play deployment used by active users.
- Partnered with 3+ clients to translate Web3 data-integrity requirements into implementation plans and coordinate feature delivery across engineering and product stakeholders in an Agile environment.

Software Engineer

PakWheels, Lahore, Pakistan -- Jun 2022 - Jan 2023

- Independently led the integration of payment gateways and supporting checkout workflows for an automotive marketplace, owning the feature from implementation through production deployment.
- Developed a Python/XGBoost vehicle-pricing model using used-car listing attributes to generate data-driven price recommendations for marketplace users.

Research and Technical Skills

Research Methods: Experimental design, human-subjects research, mixed-methods research, controlled and repeated-measures studies, pilot studies, focus groups, semi-structured interviews, behavioral observation, usability evaluation

Quantitative and Qualitative Analysis: ANOVA, Friedman and Wilcoxon tests, bootstrapping, effect sizes, clustering, eye tracking, gaze-object mapping, joint attention, gaze entropy, speech transcription and analysis, multimodal data synchronization, thematic analysis, qualitative coding

XR Prototyping and Experimentation: Unity, C#, OpenXR, Microsoft HoloLens 2, Galaxy XR, Android XR, MRTK, XR Interaction Toolkit, networked multi-user XR, Mirror, Unity Netcode, Photon Fusion, Figma

Programming and Data Tools: Python, C#, C++, JavaScript, Node.js, React, Git, Pandas, NumPy, SQL, Ruby

Teaching Experience

Graduate Teaching Assistant

Virginia Tech -- Dec 2024 - Dec 2025

- Courses: CS 1054: Intro to Programming in Java, CS/ECE 5590: System and Software Security, CS 5024: Ethics and Professionalism in Computer Science

Undergraduate Teaching Assistant

LUMS -- Aug 2022 - May 2023

- Courses: CS 360: Software Engineering, CS 5310: Computer Vision Fundamentals

Honors, Leadership and Service

- Awards and Fellowships: John Lee Pratt Fellowship (2026-2027) - Two Commonwealth Cyber Initiative (CCI) Cyber Innovation Scholar Awards (\$3,500, 2024-2026).
- Peer Review and Professional Service: Reviewer, IEEE VR (2026) and Human-Centric Intelligent Systems (2026) - Organizing Committee, iXR Ideathon (2026) - Student Volunteer, IEEE ISMAR (2025).
- Mentorship: Mentored two Virginia Tech undergraduate researchers; Computing Research Association (CRA) UR2PhD Certified Graduate Mentor (2026).
- Leadership and Honors: Graduate Council Officer, Virginia Tech CS (2026-2027) - General Secretary, LUMS Data Science Society (2021-2022) - Dean's Honor List, LUMS (2019-2022).